

HACKERTHON 2026

Full-Stack E-Commerce System (SDLC-Based Development Challenge)

9:00 AM – 5:00 PM (Offline Hackathon)

Teams must manage time effectively and follow the SDLC process step-by-step.

Hackerthon Theme

Build a Structured, Scalable, and Secure E-Commerce Web Application

Teams must demonstrate:

- Planning
- System Design
- Database Modeling
- Full Stack Development
- Testing
- Deployment (Local Execution)

Objective

Participants must:

- ✓ Apply Software Development Life Cycle (SDLC) concepts
- ✓ Design professional system diagrams
- ✓ Build a working full-stack web application
- ✓ Demonstrate frontend-backend-database integration
- ✓ Follow structured development practices

Allowed Technologies

Participants may use:

- MERN Stack

- Django + React
- Django (Full Stack)
- Node.js + Any Frontend
- Any modern full-stack framework

⚠ The technology choice must be clearly explained during demo.

SYSTEM REQUIREMENTS

Phase 1 – Planning & Requirement Analysis

Teams must prepare:

A. System Actors

- Admin
- User / Buyer

B. Functional Requirements

Examples:

- User registration & login
- Admin product management
- Category management
- Cart & checkout
- Order tracking

C. Non-Functional Requirements

- Security (Password hashing)
- Performance (Efficient queries)
- Usability (Simple UI)
- Scalability consideration
- Error handling

Phase 2 – System Design

A. Data Flow Diagram (DFD)

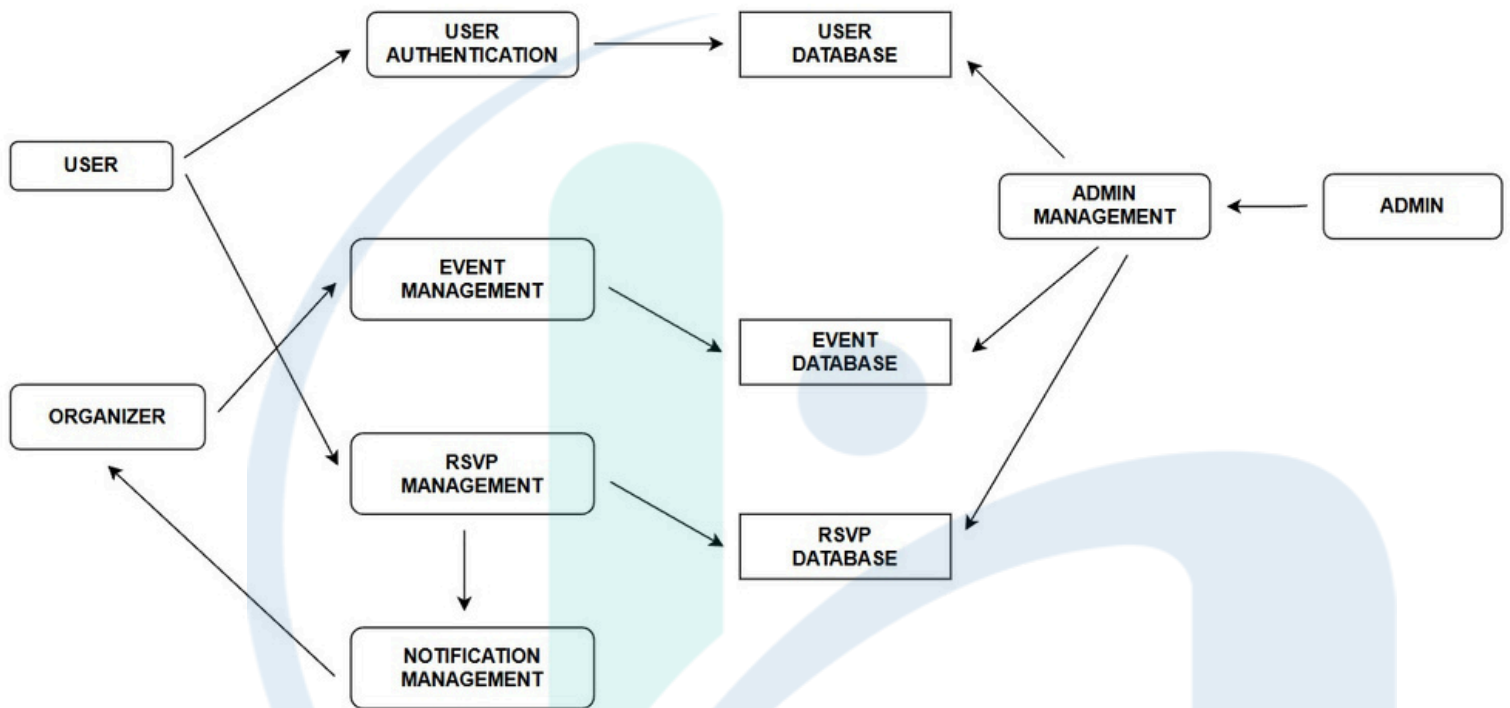
Participants must create:

✓ DFD Level 0 (Context Diagram)

✓ DFD Level 1 including:

- Authentication flow
- Category-based product browsing
- Cart flow
- Order placement flow
- Admin management flow

example:



B. Database Design (ER Diagram)

Mandatory Entities:

- User
- Admin
- Category
- Product
- Cart
- CartItem
- Order
- OrderItem

Phase 3 – Development

Develop strictly according to design documents.

MODULES & FEATURES

Admin Module

Admin Functionalities:

- ✓ Secure Admin Login (Role-Based Access)
- ✓ Add Category
- ✓ Edit/Delete Category
- ✓ Add Product under a Category
- ✓ Edit/Delete Product
- ✓ Mark product as In Stock / Out of Stock
- ✓ View All Orders
- ✓ Track Order Status (Pending / Shipped / Delivered)
- ✓ Update Order Status

⚠ Admin routes must be protected.

User / Buyer Module

User Functionalities:

- ✓ User Registration
- ✓ Secure Login
- ✓ Browse Products
- ✓ Filter Products by Category
- ✓ Search Products (Bonus)
- ✓ View Product Details
- ✓ Add to Cart
- ✓ Update Cart Quantity
- ✓ Remove Item from Cart
- ✓ Checkout
- ✓ View Order History
- ✓ View Order Status

Mandatory Core Features

Authentication

- Password hashing
- Session or JWT-based authentication
- Role-based access control

- Logout functionality

Category System

- Products must belong to categories
- Users can filter products by category
- Admin must manage categories

Product Management

- Name
- Price
- Image (URL allowed)
- Description
- Stock status
- Category relationship

Cart System

- Add items
- Update quantity
- Remove items
- Calculate total price dynamically

Order Management

- Checkout from cart
- Store order items separately
- Admin can track & update order status
- User can view order history

Phase 4 – Testing (3:30 – 4:15 PM)

Teams must:

✓ Perform manual testing

✓ Test edge cases:

- Invalid login
- Empty cart checkout
- Unauthorized admin access
- Out-of-stock product add attempt

Output:

Test Case Document including:

- Test Case ID
- Description
- Expected Result
- Actual Result
- Status (Pass/Fail)

Final Demo

Teams must demonstrate:

- SDLC process followed
- DFD explanation
- ER diagram explanation
- Application architecture
- Full working demo

Final Deliverables

Each team must submit:

- ✓ Complete source code
- ✓ Requirement document
- ✓ DFD diagrams
- ✓ ER diagram
- ✓ Test case document
- ✓ Working demo

IROHUB

Color your world with *handmade magic*

Personalized art pieces crafted to reflect your story and style.

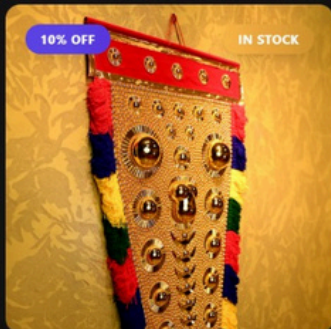


COLLECTION

Nettipattam

10% OFF

IN STOCK



1 foot Nettipattam

₹1000
₹899

"A finely handcrafted 1 foot

40% OFF

IN STOCK



Krishna and
Nettipattam fusion.

₹1000
₹600

"This Krishna and Nettipattam fusion

SHOPPING CART

Your *Collection*

 1 Items

 Empty Cart



Krishna and Nettipattam fusion.

₹600 ~~₹1000~~

- 1 +

SUBTOTAL ₹600

Order Summary

Subtotal ₹600

Shipping FREE

Your Collection Journey

Tracking the arrival of your handcrafted treasures.

 SHIPPING DESTINATION

kochi

#ORDER_SEQGLCVYI4ZMWL

PAID

 FEBRUARY 10, 2026

TOTAL VALUATION

₹600



Krishna and Nettipattam fusion.

QUANTITY 1 • ₹600 UNIT PRICE

₹600

Order Management

Tracking the journey of every masterpiece.

▼ All Orders

PAID



#order_SEQGLCvYI4Zmw1



Krishna and Nettipattam fusion.

1 Unit - ₹600

₹600

ORDERED ON

February 10, 2026

CUSTOMER

RENJITH P S

CONTACT

9876543219

ADDRESS

kochi

TOTAL REVENUE

₹600